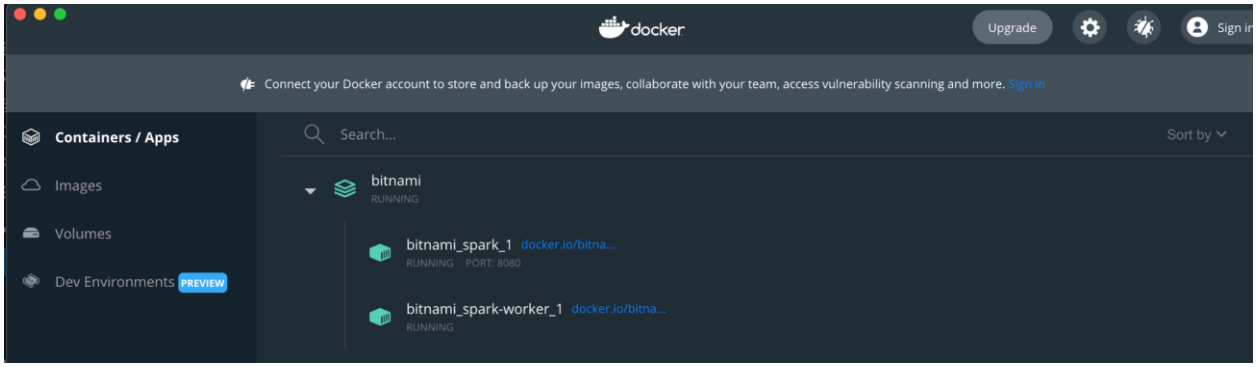
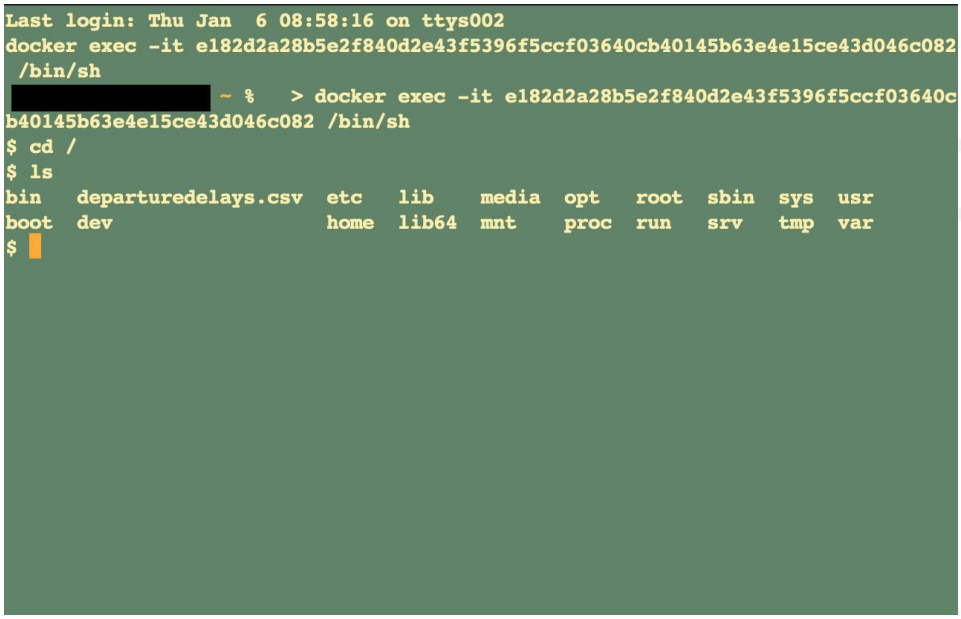
Activity 19.3 Solution File

1.



2.



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3.

```
ce43d046c082 /bin/sh
$ cd /
$ ls
bin   departuredelays.csv  etc   lib    media  opt   root  sbin  sys  usr
boot  dev                  home  lib64  mnt    proc  run   srv   tmp  var
$ pyspark
Python 3.8.12 (default, Jan  2 2022, 18:18:21)
[GCC 8.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
22/01/06 15:11:28 WARN NativeCodeLoader: Unable to load native-hadoop library for your platfor
m... using builtin-java classes where applicable
22/01/06 15:11:30 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4
041.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://e182d2a28b5e:4041
Spark context available as 'sc' (master = local[*], app id = local-1641481890704).
SparkSession available as 'spark'.
>>>
```

4.

```
$ cd /
$ ls
bin   departuredelays.csv  etc   lib    media  opt   root  sbin  sys  usr
boot  dev                  home  lib64  mnt    proc  run   srv   tmp  var
$ pyspark
Python 3.8.12 (default, Jan  2 2022, 18:18:21)
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Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://e182d2a28b5e:4041
Spark context available as 'sc' (master = local[*], app id = local-1641481890704).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>>
```

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5.

```
$ pyspark
Python 3.8.12 (default, Jan  2 2022, 18:18:21)
[GCC 8.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
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m... using builtin-java classes where applicable
22/01/06 15:11:30 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4
041.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://e182d2a28b5e:4041
Spark context available as 'sc' (master = local[*], app id = local-1641481890704).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = (SparkSession
...     .builder
...     .appName("Activity19.3")
...     .getOrCreate())
>>>
```

6.

```
Python 3.8.12 (default, Jan  2 2022, 18:18:21)
[GCC 8.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
22/01/06 15:11:28 WARN NativeCodeLoader: Unable to load native-hadoop library for your platfor
m... using builtin-java classes where applicable
22/01/06 15:11:30 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4
041.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://e182d2a28b5e:4041
Spark context available as 'sc' (master = local[*], app id = local-1641481890704).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = (SparkSession
...     .builder
...     .appName("Activity19.3")
...     .getOrCreate())
>>> activity19_3_data = "/departuredelays.csv"
>>>
```

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7.

```
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
22/01/06 15:29:27 WARN NativeCodeLoader: Unable to load native-hadoop library for your platfor
m... using builtin-java classes where applicable
22/01/06 15:29:29 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4
041.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://e182d2a28b5e:4041
Spark context available as 'sc' (master = local[*], app id = local-1641482969316).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = (SparkSession
...     .builder
...     .appName("Activity19.3")
...     .getOrCreate())
>>> activity19_3_data = "/departuredelays.csv"
>>> df = (spark.read.format("csv")
...     .option("inferSchema", "true")
...     .option("header", "true")
...     .load(activity19_3_data))
>>>
```

8.

```
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
22/01/06 15:29:27 WARN NativeCodeLoader: Unable to load native-hadoop library for your platfor
m... using builtin-java classes where applicable
22/01/06 15:29:29 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4
041.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://e182d2a28b5e:4041
Spark context available as 'sc' (master = local[*], app id = local-1641482969316).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = (SparkSession
...     .builder
...     .appName("Activity19.3")
...     .getOrCreate())
>>> activity19_3_data = "/departuredelays.csv"
>>> df = (spark.read.format("csv")
...     .option("inferSchema", "true")
...     .option("header", "true")
...     .load(activity19_3_data))
>>> df.createOrReplaceTempView("activity19_3_table")
>>>
```

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9.

```
[...     .getOrCreate())
[>>> activity19_3_data = "/departuredelays.csv"
>>> df = (spark.read.format("csv")
...     .option("inferSchema", "true")
...     .option("header", "true")
[...     .load(activity19_3_data))
[>>> df.createOrReplaceTempView("activity19_3_table")
>>> spark.sql("""SELECT date, delay, origin, destination
...     FROM activity19_3_table
...     WHERE delay > 90 AND ORIGIN = 'JFK' AND DESTINATION = 'SEA'
[...     ORDER by delay DESC""").show(10)
+-------+-----+------+-----------+
|   date|delay|origin|destination|
+-------+-----+------+-----------+
|3301755|  920|   JFK|        SEA|
|1031515|  579|   JFK|        SEA|
|1030740|  370|   JFK|        SEA|
|1041847|  220|   JFK|        SEA|
|2181505|  204|   JFK|        SEA|
|1041850|  193|   JFK|        SEA|
|1060745|  164|   JFK|        SEA|
|2031135|  142|   JFK|        SEA|
|1051515|  142|   JFK|        SEA|
|3281530|  138|   JFK|        SEA|
+-------+-----+------+-----------+
only showing top 10 rows
```

10.

```
|1031515|  579|   JFK|        SEA|
|1030740|  370|   JFK|        SEA|
|1041847|  220|   JFK|        SEA|
|2181505|  204|   JFK|        SEA|
|1041850|  193|   JFK|        SEA|
|1060745|  164|   JFK|        SEA|
|2031135|  142|   JFK|        SEA|
|1051515|  142|   JFK|        SEA|
|3281530|  138|   JFK|        SEA|
+-------+-----+------+-----------+
only showing top 10 rows

>>> spark.sql("""SELECT date, delay, origin, destination
...     FROM activity19_3_table
...     WHERE delay < 60 AND ORIGIN = 'SFO' AND DESTINATION = 'MIA'
...     ORDER by delay DESC""").show(5)
+-------+-----+------+-----------+
|   date|delay|origin|destination|
+-------+-----+------+-----------+
|1111240|   59|   SFO|        MIA|
|1062110|   58|   SFO|        MIA|
|3202105|   55|   SFO|        MIA|
|2152105|   54|   SFO|        MIA|
|2272105|   49|   SFO|        MIA|
+-------+-----+------+-----------+
only showing top 5 rows
```

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